Applied Data Mining
Assignment 1

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1. Introduction

By forecasting the following token using probability distributions obtained from internal model logits, large language models produce text. The raw logits are determined by the architecture of the model, but how these logits are transformed into actual text is determined by the decoding procedure. This generation process is greatly influenced by sampling parameters like temperature, top-k, top-p, and logit bias. Controlling originality, determinism, diversity, and dependability in AI-generated writing requires an understanding of these criteria.

This experiment explores the effects of temperature and additional sample factors on output behavior and token probability distributions. Temperature, a scaling factor applied to logits prior to softmax normalization, is the main focus. The probability distribution's sharpness can theoretically be altered by temperature. The distribution gets more predictable and sharper at low temperatures. The distribution flattens at high temperatures, enabling more varied token selection.

This work investigates the mathematical and practical implications of varying sampling parameters through controlled experiments and visualizations.

2. Methodology

2.1 Prompts Used

Prompts were chosen to allow for discernible diversity in creativity and structure in order to assess the effects of sampling settings. The purpose of the prompts was to:

- Permit unrestricted creative answers.
- Promote wording variation.
- Give structural disparities the chance to occur.
- Keep things basic enough to allow for obvious output comparisons.

Prompts for descriptive texts and short stories were two examples. These were selected because the effects of temperature are amplified by creative prompts, highlighting the distinctions between stochastic and deterministic decoding.

2.2 Temperature Values Tested

The temperature values listed below were examined:

- T = 0.0
- T = 0.3
- T = 0.7
- T = 1.0(baseline)

These settings range from extremely random generation to extremely deterministic generation. The neutral baseline with no scaling distortion is T=1.0. These numbers were chosen to symbolize progressively higher degrees of randomness:

These values were selected to represent increasing levels of randomness:

- T = 0.0 represents greedy decoding, where the model always selects the highest-probability token.
- T = 0.3 introduces minimal randomness while strongly favoring high-probability tokens.
- T = 0.7 provides moderate randomness and balances creativity with coherence.
- T = 1.0 corresponds to the unscaled softmax distribution, representing the model's default probabilistic behavior.

To guarantee a fair comparison, numerous responses were generated using the same prompts for each temperature setting.

2.3 Additional Sampling Parameters

The decoding techniques listed below were used:

- Top-k sampling (k = 3 or 4)
- Top-p sampling (nucleus; p = 0.8 or 0.9)
- Logit bias (prohibiting particular tokens)
- Combination of parameter configurations

The distribution-level effects of each approach were isolated by applying them to the same logit array.

2.4 Data Collection Process

API-based sampling with regulated parameter sets was used to create the responses. For every setup:

1. Someone gave a prompt.
2. The sampling parameters or temperature were changed.
3. There were several recorded outputs.
4. Matplotlib was used to visualize token distributions.

This made it possible to compare probability distributions and textual results directly.

3. Results

3.1 Comparison of Responses Across Temperatures

All four of the measured temperature values showed notable variations.

Greedy Decoding T = 0.0

The model consistently chose the token with the highest probability at every step when T = 0.0. This led to:

- Almost the same answers from several generations
- Extremely coherent
- Very little phrasing variation
- Strong compliance with the timely

But there was very little diversity or inventiveness. Outputs lack stylistic diversity as a result of the elimination of randomness. The most consistent responses were obtained in this environment.

Low Temperature T = 0.3

The probability distribution continued to be sharply peaked at T = 0.3. The tokens with the highest likelihood continued to dominate, but there was a slight change. Effects noted:

- A small change in wording
- Small variations in the descriptive details
- High coherence
- Low but not nonexistent inventiveness

This setting is appropriate for regulated text generation because the outputs stayed consistent and structured.

Moderate Temperature, T = 0.7

Response diversity was significantly higher at T = 0.7. The outputs varied in terms of organization, language, and stylistic choices, but coherence remained high. Effects observed:

- Enhanced richness of description
- More diverse sentence structures
- Introducing the creative components
- Decreased generational recurrence

In this experiment, this temperature offered the finest mix of predictability and inventiveness.

T = 1.0 is the baseline.

The model employed its initial softmax distribution without scaling at T = 1.0. This led to:

- Natural output diversity
- Increased fluctuation between runs
- A little bit more unpredictable
- Occasionally, there are slight structural variations

Responses were more varied but occasionally less rigidly structured than at T = 0.7. This configuration preserved adequate coherence while optimizing variation.

3.2 Temperature Distribution Visualization

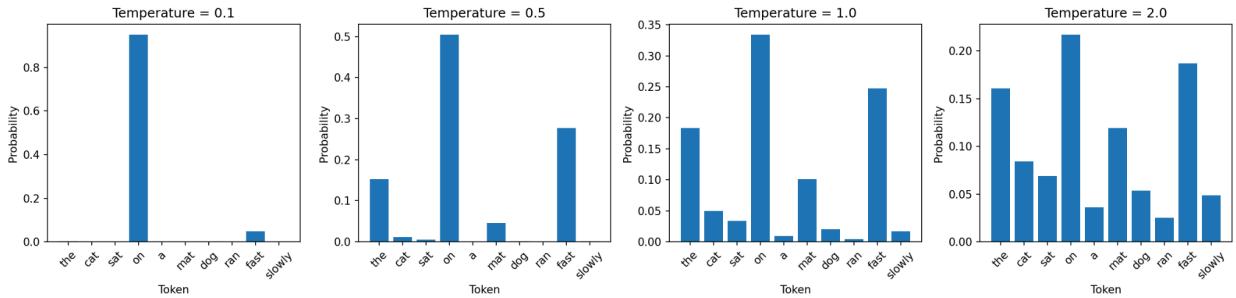


Figure 1: Effect of Temperature Scaling on Token Probability Distribution (temperature_effect.png)

The visualization makes it very evident how temperature alters probability distributions.

- Almost all probability mass focuses on a single token at T = 0.1. This happens because differences are amplified before to exponentiation in softmax when logits are divided by a tiny T.

- The distribution is still peaked at T = 0.5, but secondary tokens are still able to retain non-zero probability.

- The initial distribution is maintained at T = 1.0.

- The distribution flattens considerably at T = 2.0. The likelihood of lower-logit tokens is significantly higher than that of baseline, illustrating how entropy rises with temperature.

By altering logit ratios prior to normalization, this verifies that temperature scaling operates globally across all tokens.

3.3 Sampling Strategy Comparison

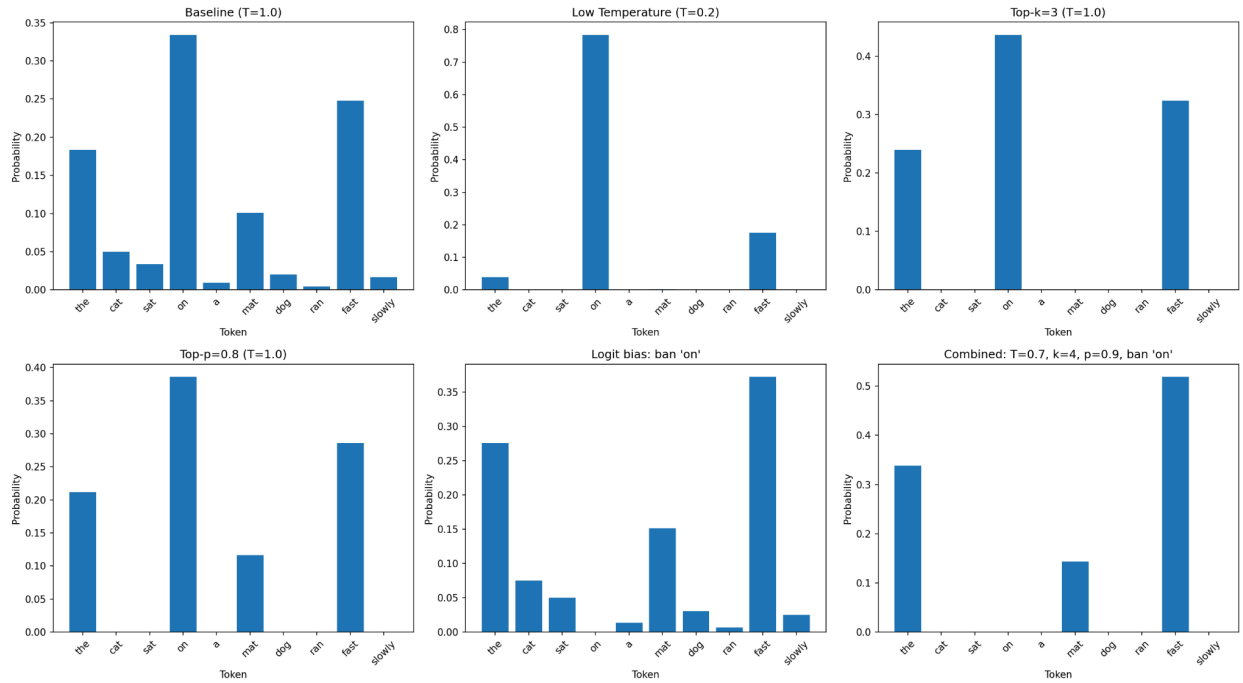


Figure 2: Comparison of Sampling Strategies on the Same Logit Array (sampling_comparison.png)

Baseline, low temperature, top-k, top-p, logit bias, and combined methods are all compared in this figure.

1. *Baseline*

According to softmax, every token maintains its probability mass.

2. *Low Temperature*

One dominant token causes the distribution to collapse.

3. *The top-k*

There are just the top k tokens left. Probabilities are renormalized and all others are clipped.

4. *The top-p*

Until the cumulative probability hits the threshold p, tokens are chosen. The quantity of tokens kept changes on the fly.

5. *Bias in Logit*

The probability mass of a banned token is redistributed, and it receives zero probability.

6. *Combined Approach*

A highly limited distribution results from applying temperature, top-k, top-p, and bias all at once. This illustrates the interplay between several sample controls.

4. Discussion

Several trends in temperature and sample techniques were found. Low temperatures decrease diversity but improve consistency and predictability. This is perfect for deterministic jobs like creating code or responding to factual questions. Although a high temperature fosters more creative variety, it can also add irrelevant or incoherent information. This helps with narrative or ideation. By introducing a strict cutoff, Top-k stops low-probability tokens from being chosen. It guarantees regulated unpredictability. Top-p is more flexible than top-k since it dynamically adjusts to the distribution shape. Targeted control over particular tokens is made possible by logit bias, which is helpful for imposing content restrictions. Temperature had the most significant worldwide impact of all the parameters. By scaling logit differences, temperature changes the entire distribution, in contrast to top-k or top-p, which impose structural restrictions.

5. Sampling Parameter Analysis

5.1 Logits and Softmax

The model produces logits, which are unprocessed, raw scores for every vocabulary token. It is not possible to directly interpret these values as probabilities. Logits are transformed into a probability distribution via the softmax function:

$$P(i) = \frac{e^{z_i}}{\sum_{j=1}^{V} e^{z_j}}$$

Softmax guarantees that probabilities stay positive and add up to 1.

5.2 Mathematical Effect of Temperature

Logits are altered by temperature prior to softmax:

$$P_T(i) = \frac{e^{z_i/T}}{\sum_{j=1}^{V} e^{z_j/T}}$$

Logit differences are exacerbated when T < 1. The probability ratio grows exponentially. Dominant tokens become overwhelmingly likely as a result.

Logit differences are compressed when T > 1. A flatter distribution results from ratios shrinking.

Figure 1 illustrates how big temperature values disperse probability more evenly, but small values collapse probability mass into a single token. The softmax function's exponential sensitivity is the cause of this.

5.3 Top-k vs Top-p Sampling

Top-k eliminates the remaining tokens after choosing the k tokens with the highest likelihood. It guarantees a candidate pool of a set size.

The smallest set of tokens whose cumulative probability is greater than Top-p chooses p. The pool's size fluctuates.

Trade-offs:

- Top-k provides a consistent constraint.
- Top-p adjusts to the distribution shape more effectively.

When rigorous control is needed, top-k is chosen; top-p is favored for balanced creativity.

5.4 Use Case for Logit Bias

In real-world moderation systems, logit bias is helpful. For instance, safe output generation is ensured by prohibiting hazardous or prohibited tokens. Additionally, it can enforce formatting restrictions, such as banning specific words from professional writing systems.

5.5 Link to API Specifications

The logit bias, top-k, top-p, and temperature scaling manual implementations directly match the API parameters utilized in production systems. It becomes clearer how API-level changes impact token selection when one understands the mathematical implementation.

5.6 Effective Top-k Intricacy

Instead of utilizing full sorting, which takes O(V log V) time, selection algorithms can be used to choose the top-k in O(V) time. A vocabulary size of V = 1,000,000 tokens makes O(V log V) computationally costly. In large-scale systems, efficient top-k greatly enhances performance.

6. Conclusion

This experiment shows that sampling parameters and temperature have a big impact on language model output. As a global scaling mechanism, temperature regulates entropy and distribution sharpness. While high temperature promotes variability at the expense of coherence, low temperature produces deterministic, predictable language.

Sampling is subject to structural limitations from top-k and top-p, each of which offers a unique balance between control and flexibility. For formatting and safety, logit bias offers targeted manipulation.

Since temperature directly alters logit ratios prior to exponentiation, it has the most significant worldwide effect overall. Low temperatures are advised for factual tasks. The ideal balance for creative writing is achieved by combining top-p sampling with temperatures ranging from mild to high.

Entropy tracking, dynamic temperature scheduling, and larger vocabulary simulations may be investigated in future studies.

7. Reflection

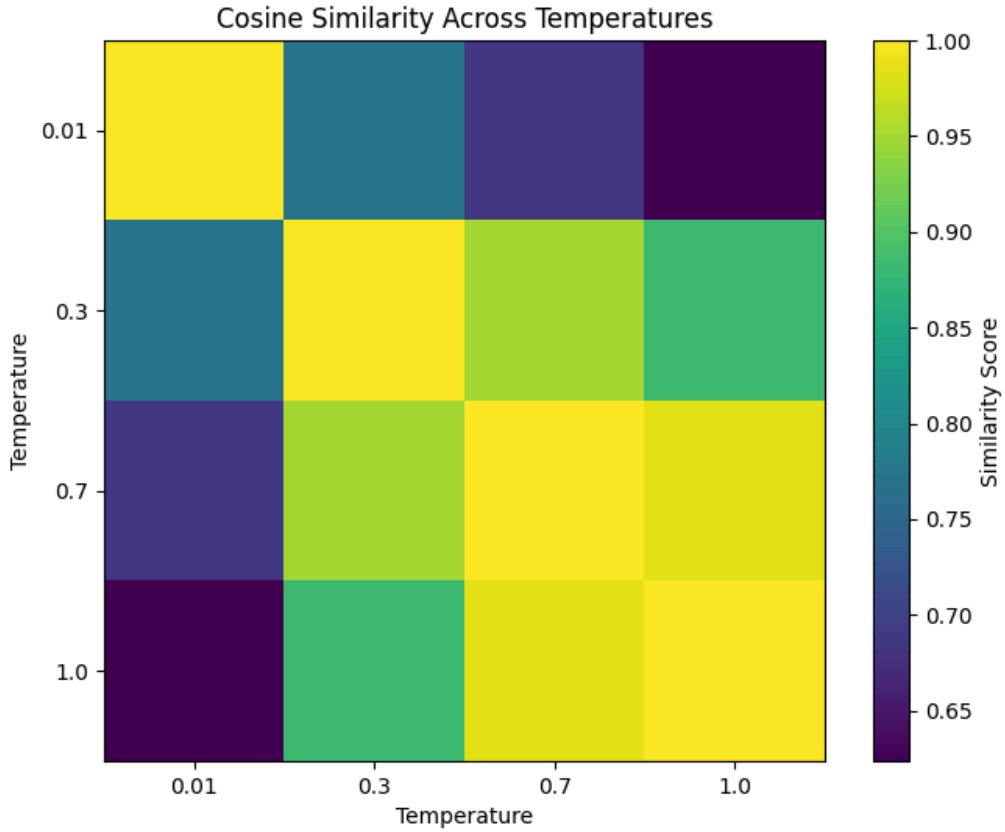
My comprehension of the core workings of API parameters like temperature, top-k, top-p, and logit bias has greatly increased as a result of implementing sampling techniques from scratch. I had a cursory understanding of these characteristics before to this project; for instance, I knew that temperature "controls randomness." But after implementing softmax and manually adding

temperature scaling to logits, I discovered that temperature mathematically rescales logit differences before exponentiation rather than randomly injecting noise. Observing the probability ratio formula $e^{(z_i - z_k)/T}$ demonstrated how token dominance can be compressed or amplified exponentially by slight temperature changes. In a similar vein, applying top-k and top-p sampling showed that these techniques are accurate truncation processes that modify the distribution before renormalization rather than nebulous "filters."API parameters were converted from abstract settings into concrete mathematical operations when I wrote the code.

The severity of the effect of T = 0.0 was one finding that caught me off guard. At first, I thought it would only marginally lessen unpredictability, but in reality, it eliminated stochastic behavior and limited sampling to argmax selection. With probability 1, the distribution essentially collapsed to a single token. As a result, the model produced outcomes that were almost the same from run to run. The degree to which modest adjustments, like going from T = 0.3 to T = 0.7, improved output diversity without seriously impairing coherence also caught me off guard. In actuality, the difference felt considerably greater than I had predicted in theory. Overall, this experiment showed that language model behavior can be disproportionately affected by tiny parameter changes.

8. Extra Credit

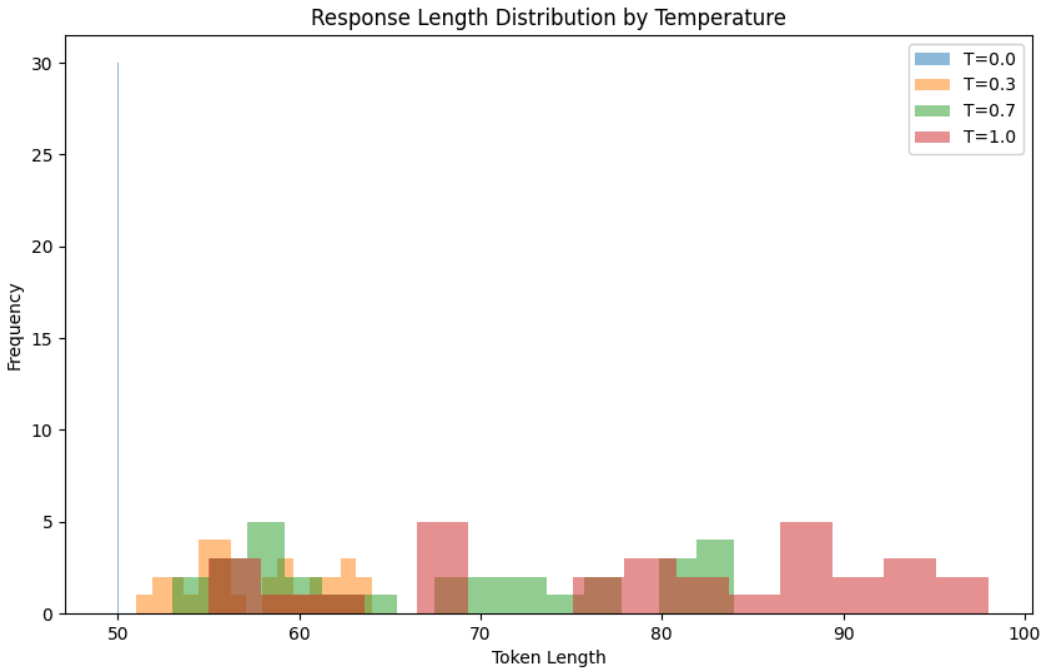
8.1 Cosine Similarity Across Temperatures



This matrix quantifies the degree to which temperature changes alter the probability distribution.

- The similarity scores are extremely high (around 1.0) at low temperatures (near 0). The AI becomes "deterministic," consistently selecting only the most likely word, which explains this.
- The similarity decreases at high temperatures (about 1.0). The dispersed distribution indicates that the AI is taking into account a large number of word choices, producing more varied and imaginative (albeit occasionally less cohesive) outcomes.
- Important Takeaway: Higher T produces distinct, variable patterns, while lower T produces consistent, predictable patterns.

8.2 Response Length Distribution



This histogram illustrates how the AI's "predictability" influences the length of its responses.

- Pay attention to the tall, thin bars at T = 0.0. The AI follows a strict path at zero temperature, which frequently produces consistently very comparable response lengths.

- The bars are shorter and more widely spaced along the x-axis when T = 1.0 is present. This is what is meant by "variance." Some courses lead to extremely short responses, while others veer into much longer ones since the AI is selecting more random words.

- Key Takeaway: Low temperatures maintain responses within a small, predictable range, but high temperatures increase the output length's uncertainty.